

Question	Answer	Marks	Guidance
6(a)	work done per unit charge	B1	Ratio must be clear. Allow 'energy' for 'work done'. Ignore any symbols unless they are defined. Ignore any reference to units. Ignore $V = Q / 4\pi\epsilon_0 r$.
	work done (on charge) in moving positive charge from infinity (to the point)	B1	Allow any combination of positive/negative, on/by and from/to infinity that gives a positive sign. E.g. 'work done on negative charge in moving to infinity', 'work done by negative charge in moving from infinity' etc.
6(b)(i)	$V_0 = Q / 4\pi\epsilon_0 R$, where ϵ_0 is the permittivity of free space	B1	Ignore sign of answer.
6(b)(ii)	curve from $x = R$ to $x = 3R$ with decreasing magnitudes of V and gradient, starting at (R, V_0)	B1	Ignore anything drawn in the $x = 0$ to $x = R$ region. If curves are drawn in both + and – regions, both curves must meet the requirements for MP1 & MP2. Allow tolerance $\pm\frac{1}{2}$ small square horizontally. Allow this mark if curve starts at $(R, -V_0)$. Line may become horizontal at the end for no more than 4 small squares, and must not show an increasing magnitude of more than $\frac{1}{2}$ small square.
	line passing through $(2R, \frac{1}{2}V_0)$ and $(3R, \frac{1}{3}V_0)$	B1	Allow tolerance $\pm\frac{1}{2}$ small square vertically. For the V value at $3R$, expect the line to pass on or above the third minor division above the x -axis. Allow this mark for $(2R, -\frac{1}{2}V_0)$ and $(3R, -\frac{1}{3}V_0)$.
	line entirely within the positive V region	B1	Allow any shape for this mark, but do not allow if anything is drawn in the negative V region. Allow curve to touch the x -axis but not cross it.
6(c)(i)	$Q = (-)4.80 \times 10^4 \times 4\pi \times 8.85 \times 10^{-12} \times 0.119$	C1	Allow C1 without the minus sign. Allow '1 / (8.99 $\times 10^9$)' for '4 $\pi \times 8.85 \times 10^{-12}$ '.
	$= -6.35 \times 10^{-7} \text{ C}$	A1	Correct to at least 3 significant figures. AFC. (6.352) Answer must be negative for the A1 mark.

Question	Answer	Marks	Guidance
6(c)(ii)	stores energy / charge	B1	Allow 'Can be charged / discharged'.
6(c)(iii)	$C = Q / V$	C1	Allow $C = 4\pi\epsilon_0 R$.
	$C = (6.35 \times 10^{-7}) / (4.80 \times 10^4)$ $= 1.32 \times 10^{-11} \text{ F}$	A1	Q / V method: Possible ECF from answer in 6(c)(i). $4\pi\epsilon_0 R$ method: Possible ECF from R value from 6(c)(i) with if a POT or TE error that led directly to a mark not being awarded is repeated. Correct to at least 3 significant figures. AFC. (1.323) Allow 13.2 pF provided unit is clear. Unit needed with answer for A1 mark. Allow C V^{-1}, $\text{s } \Omega^{-1}$, $\text{A}^2 \text{ kg}^{-1} \text{ m}^{-2} \text{ s}^4$.
6(c)(iv)	$E = \frac{1}{2}QV$ or $E = \frac{1}{2}CV^2$ or $E = Q^2 / 2C$	C1	
	$= \frac{1}{2} \times 6.35 \times 10^{-7} \times 4.80 \times 10^4$ $= 0.0152 \text{ J}$	A1	Possible ECF from 6(c)(i) and/or 6(c)(iii). Correct to at least 2 significant figures. AFC. Treat AFC answer as 0.015 J.